

Structural vulnerability in transmission systems: Cases of Colombia and Spain



Gabriel J. Correa^a, José M. Yusta^{b,*}

^a Faculty of Engineering, Fundación Universitaria Luis Amigó, Medellín, Colombia

^b Department of Electrical Engineering, Universidad de Zaragoza, Zaragoza, Spain

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ABSTRACT

In this paper the authors apply methodological strategies for the structural vulnerability assessment in high voltage power networks based upon the combination of power flow models and scale-free graph statistic indexes. Thus, it is possible to study risk scenarios based on events that may trigger cascading failures within a power system. The usefulness of graph theory techniques has been validated on previous works of the authors, and may be applied in analysis of the vulnerability of different power electric networks. A case study for vulnerability analysis is carried out through methodologies that allow comparison on random error and deliberate attack tolerance evaluation in transmission electric systems from countries like Colombia and Spain. Such vulnerability assessment methodology takes into account the current conditions of the power networks (base case), as well as the impact of expansion plans into infrastructures as defined by their governments. Consequently, the authors show the advantage on the use of graph theory based techniques for vulnerability analysis of electrical power systems.

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1. Introduction

State governments nowadays consider security of supply as one of the principal objectives within their energy policies, involving institutional efforts on the protection of homeland security, economic activities, public health, etc. This fact evidences the close relationship between global security of electric infrastructures with other critical infrastructure sectors of the economy [1,2]. This concern has been addressed by both European Commission and US Homeland Security Department regarding the security of countries' infrastructures, as a response against new international threats. In 2008 the European Council adopted Directive 2008/114/CE [3], which gave rise to the "European Programme for Critical Infrastructure Protection (EPCIP)". In 2009, the National Infrastructure Protection Plan (NIPP) [4] was launched in the United States, following the release of several frameworks established by the US Department of Homeland Security [5,6]. In order to achieve the goal of energy infrastructure protection, policies suggest implementation of risk management programmes involving vulnerability analysis, threat assessments and risk control measures [7]. The risk management framework involves dealing with sets of risks that are most likely to impact critical infrastructures

operations, ranging from severe weather conditions and technical failures in assets, to sabotages and terrorist attacks.

Several episodes of sudden disruptions in the electrical service covering large geographic areas have deeply impacted both economic and social sectors in affected countries. Some resembled blackout incidents refer to events on Canada and the U.S. (August 2003), Germany, Belgium, Italy, France, Spain and the Netherlands (November 2006), Brazil (November 2009), as well as short-term acts of terrorism on certain assets of electric infrastructures, such as those experienced in Colombia (1998–2003). As a result, the need to incorporate studies to assess vulnerability of systems has become a requirement in the management of those risks that impact the operations of the electric infrastructures.

Power grid vulnerability, either locally or nationally, is usually tight to the interest of the system operating companies. Most of vulnerability studies are carried out after the occurrence of high-impact events (for example, a widespread blackout) determining the causes of cascade failure events within a specific power grid. Such studies are achieved through structural vulnerability analysis in power transmission networks, requiring well-established methodologies that may guide decision-makers on prevention and recovery from disruptions on the power grid. For example, N-1 and N-t contingency studies are among the most used criteria in power industry [8,9].

Graph theory constitutes a relatively new knowledge area in which it is possible to study interdependencies among critical infrastructure systems, specifically the power electric network. In

* Corresponding author. Address: Universidad de Zaragoza, Departamento Ingeniería Eléctrica, C/Maria de Luna 3, Zaragoza 50018, Spain. Tel.: +34 976761922; fax: +34 976762226.

E-mail addresses: gabriel.correa@amigo.edu.co (G.J. Correa), jmyusta@unizar.es (J.M. Yusta).

this paper, the authors apply a method based on graph theory modelling (theory of complex networks), which allows a simpler representation of electric infrastructures, by analysing their robustness when disrupted. This implies the study of vulnerability due to effects of certain risks and threats that affect normal operation of the power grid [10].

Modelling of power systems as complex networks has focused important research in the last few years, following the first definition of scale-free networks proposed by Barabási and Albert in [11]. Such definition was followed by the release of concepts on statistical measures, vulnerability analysis, robustness and resilience estimation in scale-free networks, with applications to power grids, computer networks and many other engineering areas [10,12,13]. Motter and Lai in [14] formulated the problem of cascading failures into scale-free networks by modelling a degree-based node removal strategy, which constituted the basis to perform vulnerability assessment into electric power networks [15]. As discussed by Holmgren in [16,17] graph theory statistical measures (clustering, connection degree, geodesic distance, nodal distribution) are suitable to perform vulnerability assessment of a power system and for estimation of events that may trigger cascading failures.

Other measures obtained from complex networks models are also suitable to perform vulnerability assessment of electric power networks, as formulated by Chen et al. in [18], who have suggested the use of geodesic efficiency indicators to diagnose cascade event failures. Other approach to assess the power grid's condition on robustness and resilience is formulated by Johansson in [19] and by Wang et al. in [20] who propose measuring the betweenness among nodes of the graph model.

Among many graph models of the electric power grid in certain countries, it is worthy to bring up some studies performed upon the system of the Nordic countries [16,19,21], the continental European power system [22,23] and some South American countries [24], in which are shown advantages of calculating topological measures (node clustering, geodesic distance, geodesic efficiency) to analyse the grid's vulnerability.

So far, modelling with complex networks has become a well-accepted methodology to perform studies for protection of electric networks against certain threats, as revealed by Holmgren et al. in [25] and by Chen et al. in [18,26,27]. Furthermore, a recent research formulated in Correa et al. [28,29] has provided a validation of the equivalence between power load flow techniques and scale-free graph modelling in order to perform vulnerability studies in transmission networks. This fact allows assessing the vulnerability of electric infrastructures based on graph theory models, which are a least time consuming technique, instead of classic power flow indexes.

Based upon such innovative contribution, in this paper the authors apply a modelling graph theory methodology in order to perform a comparative vulnerability assessment on the electric power systems of two countries (Colombia and Spain). This includes an analysis of the network's tolerance against random errors and against deliberate attacks. Moreover, an analysis of the effectiveness of expansion investments is performed according to the information established in each country's power grid expansion planning.

The paper is organized as follows: Section 2 introduces the scale-free topology equivalence for power networks and describes appropriate indexes to measure power grid disruption events. Section 3 shows a case study according to an algorithm for random errors and deliberate attack vulnerability assessment applied to some illustrative examples that are based upon the topologic representation of power networks in Colombia and Spain. This corresponds to a practical application of the methodology previously validated in [29]. Discussion and conclusions on usefulness of

scale-free graph modelling and vulnerability assessment is also provided at the end of the paper.

2. Modelling electric power systems with complex networks

It is necessary to transcend risk qualitative assessment in order to generate an effective evaluation framework of vulnerability in transmission, which must involve the study of relations with other systems, their interdependence and their response to both threats and changes under different operating conditions. This involves the study of the events that trigger cascade failures and disconnection of consumers. The methodology explained in this section could be particularly useful as a mechanism for explaining events like power outages or blackouts.

2.1. Proposal for representing network topology

Electric networks can be resembled to scale-free graphs [11], which allows the representation of most assets that conform the power grid. Some researchers simplify such representation as a complex network where substations are specified as *nodes* and electric lines are sketched as *links* [16,19,21,26,30]. In those cases such approach allows the calculation of cluster measures (triangles) in order to determine the grid's vulnerability.

Such representation is quite simple, but incomplete, since many important assets in the power grid cannot be properly taken into account in structural vulnerability assessment, as they are, transmission towers, transformers, generation centres, load consumers, capacitors and other assets.

Fig. 1 shows the proposed topological representation of a 5-bus electric network, compared to the traditional representation (which only considers buses and links). Note that both transformers and electric towers are also taken into account as assets susceptible to be removed due to attacks or errors in the power grid. Thus, the resulting network is constituted by a graph of 16 nodes and 17 links. The topological representation herein proposed looks for a more realistic representation of the power system as a scale-free graph, where the set of towers that hold power lines are also considered as a node in the graph. Similar consideration is made for the set of transformers.

In scale-free graphs few nodes are highly connected, meaning that they have a large number of edges to other nodes, although the degree of connection throughout the graph is quite low. Such graphs are closer to reality, since the network will grow preferentially on the basis of the nodes of greater connectivity (preferential attachment) [11,12].

2.2. Graph theory and power flow indexes

This section shows some statistical measures that describe scale-free graphs in order to analyse the disintegration of networks (evolution to successive node removal). The formulation of these measures are based upon the definition of the *geodesic distance* d_{ij} concept which describes the shortest distance between two nodes directly, by counting the minimum number of traversing nodes required to join them [31].

2.2.1. Geodesic efficiency

This indicator was originally proposed by Latora and Marchiori in [32], for estimation of the efficiency in which information is exchanged within a network. It is assumed that the flow between two nodes must pass through the shorter geodesic distance [18,26]. Therefore, the *efficiency* between a pair of nodes e_{ij} is defined as the inverse of their geodesic distance. If there were no connection

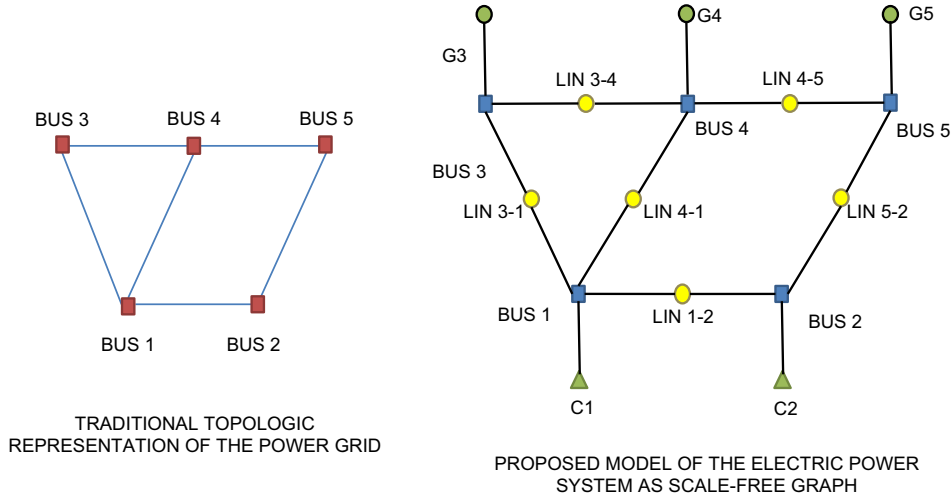


Fig. 1. Proposed representation of an electric network as a scale-free graph (5 bus network).

between two nodes, $d_{ij} \approx \infty$, $e_{ij} = 0$. It is possible to calculate the *Graph Average Efficiency* ($\bar{e}$) formulated as:

$$\bar{e} = \frac{1}{N \cdot (N-1)} \sum_{i \neq j} \frac{1}{d_{ij}} \quad (1)$$

In an electric power system, a low geodesic efficiency means that the flow of electricity must flow through a greater number of nodes, and thus may increase line's capacity problems or overloads.

2.2.2. Geodesic vulnerability index ($\bar{v}$)

The *average geodesic vulnerability index* ($\bar{v}$) is directly related to *geodesic efficiency* in Eq. (1). The proposed index allows better measurements in network's performance when subjected to contingency events, with regard to its steady condition (base case). This index also standardises geodesic efficiency and permits effective comparisons of successive iterations of cascading failures events.

$$\bar{v} = 1 - \frac{\sum_{i \neq j} \left(\frac{1}{d_{ij}^{LC}} \right)}{\sum_{i \neq j} \left(\frac{1}{d_{ij}^{BC}} \right)} \quad (2)$$

d_{ij}^{LC} is the shortest path between a pair of nodes of the scale-free graph, after a node removal in each iteration. d_{ij}^{BC} is the shortest path between a pair of nodes of the scale-free graph at base case.

Index $\bar{v}$ in (2) varies between zero and one. The higher *vulnerability* index value $\bar{v}$, the greater impact on the network due to congestion problems and cascading failures, as some geodesic paths get disrupted and therefore, electric power must flow through more paths. This is consistent with the network's fragmentation and isolation of power loads into the system. Therefore, $\bar{v}$ index can be measured as a function of the *fraction of removed nodes* (f).

Previous studies performed by Correa et al. in [29] demonstrate the effectiveness of index ($\bar{v}$) to assess power grid vulnerability.

2.2.3. Load shedding (LS)

The study of cascading failures requires the calculation of power system performance through estimation of the loads that remain served under contingency events. A comprehensive index that enables the estimation of disconnected loads under cascading failures corresponds to *load shedding LS*. The calculations proposed by Salmeron et al. [33], Motto et al. [34], Donde et al. [35] are limited to estimations of load shedding relative to the active power, whereas Bier et al. [36] and Haidar et al. [37] estimated load shedding as a percentage of the total system demand. *Load shedding*

then can be established as an estimation of the total apparent power that remains connected in cascade failures events, as formulated in Eq. (3). This index is appropriate to estimate the structural performance of the network, showing the percentage of non-supplied demand in a particular network.

$$LS = 1 - \frac{\sum_i \sqrt{(P_{Di}^{LC})^2 + (Q_{Di}^{LC})^2}}{\sum_i \sqrt{(P_{Di}^{BC})^2 + (Q_{Di}^{BC})^2}} \quad (3)$$

P_{Di}^{LC} is the active power load that remains electrically connected, after a node removal in each iteration. Q_{Di}^{LC} is the reactive power load that remains electrically connected, after a node removal in each iteration. P_{Di}^{BC} is the active power load at base case. Q_{Di}^{BC} is the reactive power load at base case.

LS index in Eq. (3) is calculated as a percentage of the total load disconnected from the grid after each removing a node and varies between zero and one. As LS index increases, the impact on the apparent power loads of the grid increases as well, measured as a function of the *fraction of removed nodes* (f).

2.2.4. Relation between LS and $\bar{v}$ indexes

There is equivalence between the results obtained using graph theory and the results of power flow techniques. The utility of combining power flow models and scale-free graphs measures both numerically and graphically has been validated in Correa et al. [29], by showing correlation between the LS index of Eq. (3) and $\bar{v}$ index of Eq. (2) in terms of events that trigger cascading failures. Such validation has been achieved by calculating the correlation coefficient ρ between indexes *load shedding LS* and *geodesic vulnerability $\bar{v}$* in different IEEE test networks (14, 24, 30, 57, 118 buses). This showed a positive linear relationship between both indexes, since ρ value is near to +1. This correlation ρ implies an important advantage when combining classic methodologies of electric power flows and graph theory measurements in order to study cascade failures events [28,29].

According to this comparison $\bar{v}$ index is useful to determine the proportion of disconnected loads during cascading failures events. Therefore the existence of a correlation may be simplified as suggested by Correa et al. in [29]:

$$\bar{v} \sim LS \quad (4)$$

Since scale-free graphs indexes can be used to estimate the vulnerability of a power grid topology compared to another one, it is possible to substitute time-consuming computational tools (as classic power

flow routines) with more efficient techniques (as graph theory parameters) to evaluate structural vulnerability of any electric network, depending on the events that trigger cascade event failures.

2.3. Network tolerance to both random errors and deliberate attacks

One of the objectives of vulnerability analysis is to assess the system conditions and track the evolution of its changes that trigger cascading failures. Concepts on disintegration of scale-free networks were first introduced by Albert and Barabási in [12], whose studies show the performance of complex networks in case of events of systematic node removals at either random events (errors tolerance) or deliberately actions (attacks tolerance). Analysing contingencies that lead to cascade failures on scale-free networks involves the use of parameters that measure the evolution on the connectivity and functionality of the network. This is performed through successive iterations that represent node removal from the network. Each node removal is associated to a contingency and considered as an iteration step in the network disintegration. A node removal also implies the elimination of all links connected to it, and therefore, the corresponding geodesic paths also disappear.

2.3.1. Removal strategy due to random errors

Tolerance to random errors relates to those damages in infrastructure systems triggered by random contingencies, for example, natural disasters, equipment failures or human error. It has been shown that this kind of cascading failures may cause the collapse of the network service (blackout) when isolated up to 20% of the system nodes [29].

2.3.2. Removal strategy due to deliberate attacks

Tolerance to deliberate attacks relates to targets in the network that are chosen deterministically by the attacker. An extreme scenario, but possible, is one in which the targets of the network are defined as most important, either by their nodal degree [16,17,21], or by their betweenness degree [19,27]. It has been shown that deliberate attacks may cause blackout of the grid when isolated 2–5% of the nodes in the system [29].

3. Case study in transmission networks

On this section, two countries' power grids are modelled as complex networks (Colombia and Spain) whose topology is defined according to their current structure. Each network's tolerance is analysed taking into account the robustness provided by their expansion plans. This case study involves the representation of assets as nodes (substations, electrical towers, transformers, connection bars, etc.) and links (airlines, underground, etc.), regardless of physical distances or the electrical parameters (line impedances, voltage regulation, power loss, etc.) in the network.

3.1. Procedure to assess structural vulnerability in transmission networks

A practical application of the proposed methodology is carried out in 400 kV power grids in Spain and 220 kV/500 kV power grids in Colombia. This includes the analysis of the expansion plans published by the ministries that are responsible for such planning. Fig. 2 shows the proposed algorithm to evaluate geodesic vulnerability $\bar{v}$ as a function of the fraction of removed nodes f from the network topology. Note that for implementation of the algorithm showed in Fig. 2 it is not required to include all the electrical parameters of the network under analysis. Therefore, the proposed methodology implies an advantage, since most information concerning the network topology is public whereas access to electrical data is very limited.

Error distribution on N-t contingency analysis is highly asymmetric due to its random nature. In these cases, the authors propose following the *Central Limit Theorem*, which suggests that a sufficiently large number of independent random values will be approximately normally distributed when the sample size is relatively large. Thus, it is possible to conclude that the approximation is good enough when more than 30 samples have been collected [38]. Structural vulnerability studies presented in subsequent sections include analysis of network tolerance against random errors, corresponding to the average of 35 samples to describe the cascading failure indexes, unlike deliberate attacks that run only one sample by the disruption of each node's graph, starting by the most connected nodes (degree-based removal).

The flow diagram in Fig. 2 has been implemented in **Matlab**[®]. The script incorporates features of the *MatlabBGL* toolbox for graph theory [39]. Geodesic distances are calculated through shortest-paths Bellman–Ford algorithm [31].

3.2. Network topology description

This section contains a brief explanation of the network topology and the relationship with its robustness. The study is applied into infrastructures that comprise the set of lines, connections and other assets operating at voltages of 220 kV and 500 kV in the case of Colombia and 400 kV in the case of Spain.

3.2.1. 400 kV Transmission network in Spain

The Spanish interconnected is composed by more than 40,000 km of power lines, over 4000 positions in substations and about 75,000 MVA transformation capacity [40]. It is possible to access public information on the topology of the high voltage network in Spain, such as that contained in the maps of the networks [41] including an outline of most assets.

In order to illustrate the application of the methodology this paper focuses on the 400 kV networks that are represented in the scale-free graph showed in Fig. 3. It is composed by 48 substations, and 70 transmission lines (set of towers plus cables) that are interconnected among them [42]. Fig. 3 contains a scale-free graph representation of the different connecting points including electrical loads and generators that inject active power into the network.

Assets in the Spanish power grid are managed by the firm *Red Eléctrica de España*, which is responsible for operating and maintaining the transmission network, both within the Iberian Peninsula, Balearic Islands and Canary Islands. This consolidates the company as the only single electrical system transporter and grid operator in Spain.

3.2.2. 220 kV/500 kV transmission network in Colombia

Near 75% of the Colombian territory is not connected to the national interconnected network. The electric grid is primarily spread through the region of the Andean highlands and on the Caribbean coast region as well as large urban centres located on the Pacific region, corresponding to the zones that focus most of population and country's economic life [43,44]. The effective power capacity installed in the Colombian grid at the end of the year 2011 was 14,420 MW, of which 9,200 MW (64%) came from hydropower [45]. Most of the hydroelectric generation centres are located in the Andean region, whereas thermal generation is mostly located in the Caribbean region and in the highlands Cundinamarca–Boyacá at the centre of the country.

Some previous works [24] and reports delivered by Colombian government [43] constitute the basis to construct an equivalent scale-free graph of the network topology as shown in Fig. 4.

Assets in the Colombian interconnected power grid are managed by the firm *ISA (Interconexión Eléctrica S.A)*, which is also responsible for their operation and maintenance (As information,

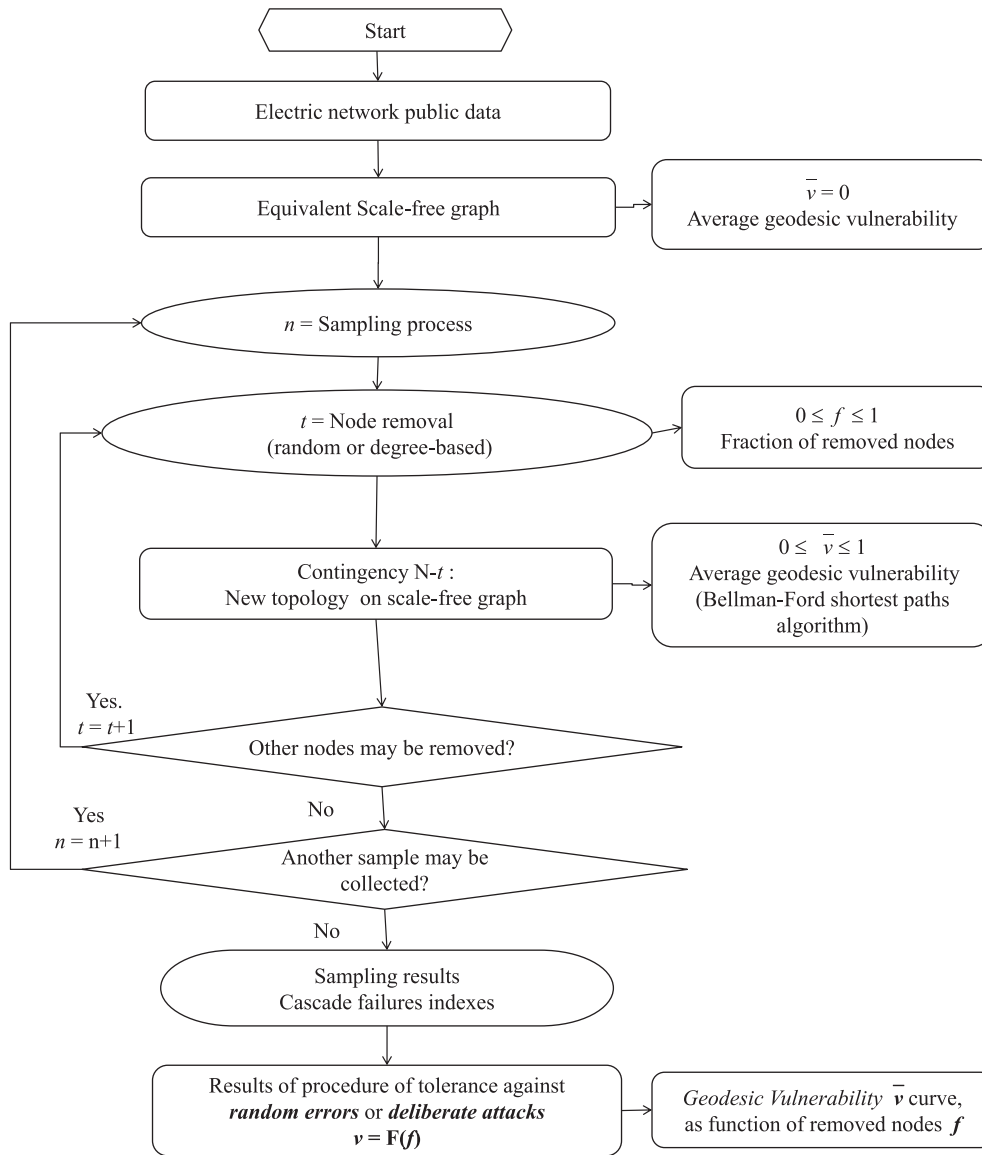


Fig. 2. Flow diagram to calculate structural vulnerability in transmission networks.

ISA is mainly owned by the Colombian government. The company is also responsible for operation of transmission networks in Brazil, Bolivia and Peru).

3.3. Case study description

Once the scale-free graph representation has been achieved, the structural vulnerability of the electric system is evaluated according to three cases in the networks of Spain and Colombia.

3.3.1. Case 1: base case network

Case 1 consists on vulnerability assessment on the current state of the network which was previously described in Section 3.2. The case takes into account the characteristics of the transmission network in Spanish Peninsula [41] as well as the features on the transmission network of the Colombian interconnected system [45].

3.3.2. Case 2: robustness improvement plans

For case 2 vulnerability assessment is performed taking into account those plans to improve robustness of the network, as provided by the respective governmental agencies. Robustness must be understood as a strategy of improvement in the mesh of the

network, by adding new connections among current buses. The case 2 only considers the effect of building new transmission lines, without adding new substations.

The planning document of the Spanish network for the 2008–2016 period [46] aims to strengthen the meshing of the 400 kV network between those nearest substation. Thus, the plan seeks to increase system reliability and to ensure dispatch of the power generated at wind farms that have being built throughout different autonomous communities.

In Colombia, for the period 2010–2024 [43], robustness planning of 220 kV and 500 kV transmission grid intends to consolidate the network mesh in order to make it more robust against deliberate attacks and specific failures, especially relating the interconnection between Andean and Caribbean region.

3.3.3. Case 3: expansion plans

Case 3 consists on assessing vulnerability of the expansion plans of the existing network, as stipulated in the planning time-periods of transmission networks. The case 3 considers the effect of building new transmission lines as well as the addition of new substations in the electric network. This leads to the evaluation

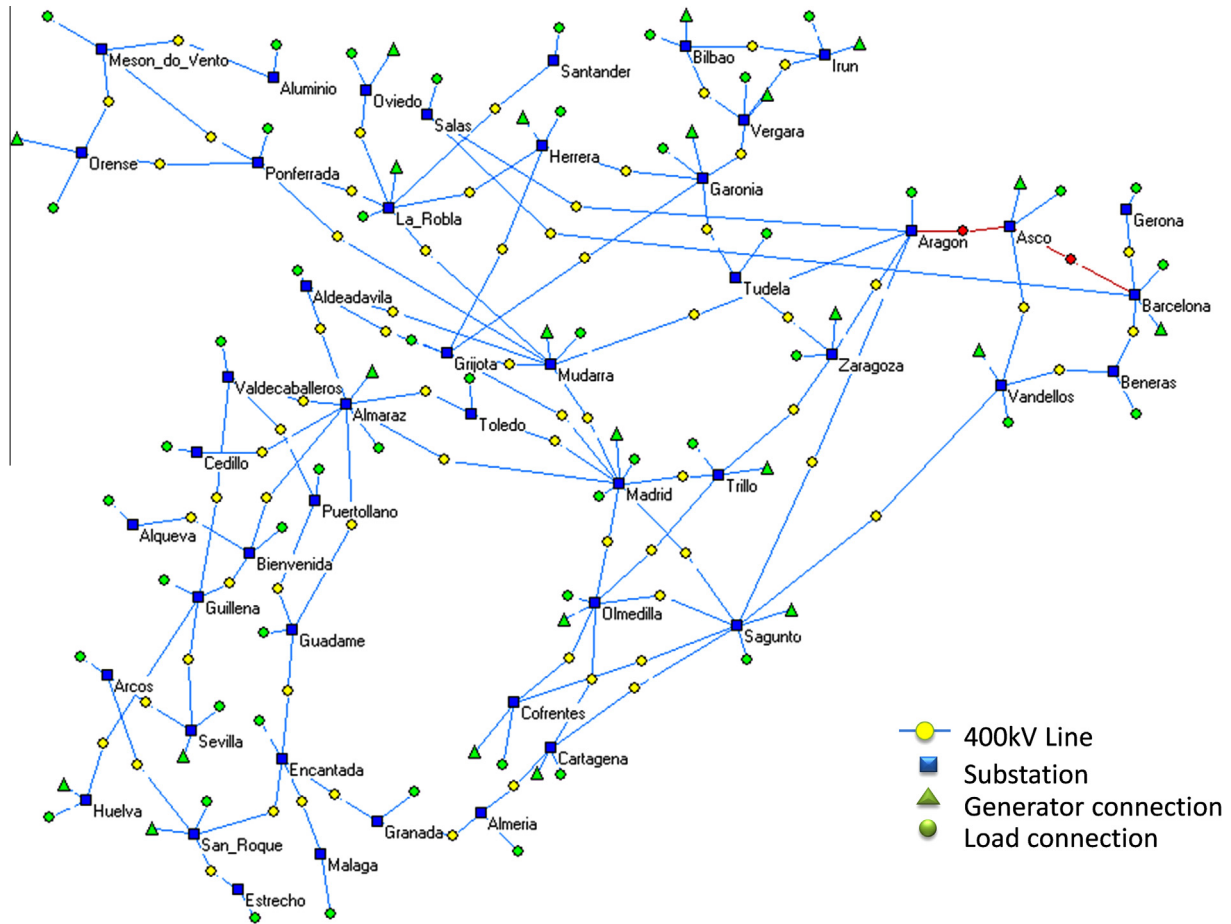


Fig. 3. Scale free graph of 400 kV transmission network in Spain.

of a new grid topology according to the indicative plans expansion of each country.

Planning the expansion of the transmission network in Spain emphasizes on connecting new wind farms generation and new gas plants to the 400 kV grid during the next years up to 2016 [46]. The connections of large electric loads, such as high speed trains, have been taken into account in the projection of the new power lines throughout the country.

The Colombian planning expansion 2016–2024 seeks to install new substations in geographical locations bordering neighbouring countries (Panama, Ecuador, Venezuela) and connect new hydro-power for future electric interchanges. Additionally, with the rise of new development centres in the Orinoco region it is necessary to improve meshing of the grid in those regions where demand for electricity will keep growing.

Table 1 shows the quantities of buses, nodes and links involved in the scale-free graph modelling, according to the case study descriptions given in this section, in order to perform further vulnerability assessment. Moreover, calculations are provided for both graph's average nodal degree ($\bar{k}$) and graph's average efficiency ($\bar{e}$) as previously stated in Eq. (1).

3.4. Structural vulnerability responses

Estimating the structural vulnerability of networks involves tolerance evaluation of the network against both random errors and deliberate attacks, according to the algorithm previously specified in Section 3.1. Vulnerability analysis is then performed based upon the fraction of removed nodes f , which is associated to certain risk

scenarios, for example, natural disasters on which certain percentage of the territory becomes isolated.

The implementation of the algorithm presented in this section was performed on a personal computer with *Matlab*® version 7.2, and hardware corresponding to an *Intel Core Duo* 2.33 GHz processor, and 2 GB of RAM memory. Table 2 shows some statistics relevant to the simulation of deliberate attacks and random errors on the networks. Note that the number of iterations per sample is greater in random disruptions than node degree based attacks.

Each sample contains the results of successive iterations for node removal after executing contingency $N-t$ in cascade failure analysis, including the base case.

3.4.1. Nodal degree distribution and scale free graph indexes

Nodal degree distribution reveals how complex networks are arranged. The representation as scale-free graphs implies that certain nodes with many connections become cornerstones of the network. Fig. 5 shows the nodal degree distribution $P(k)$ for 220 kV/500 kV Colombian network and for 400 kV Spanish network. It also includes a comparison for nodal degree distributions among cases 2 and 3 (described in 3.2.1 and 3.2.2), according to the network expansion plans [43,46]. These distributions have been calculated by means of *MatlabBGL* toolbox for graphs [39].

Nodal degree distribution $P(k)$ presented in Fig. 5 specifies the probability that a randomly chosen node from the network has exactly k connections (or neighbours). As showed in Fig. 5, in the base case 1 for the Spanish network, the probability that a node in the graph has a degree $k=2$ is $P(k)=0.4$, which means that 40% of the grid nodes have only two connections to other network nodes.

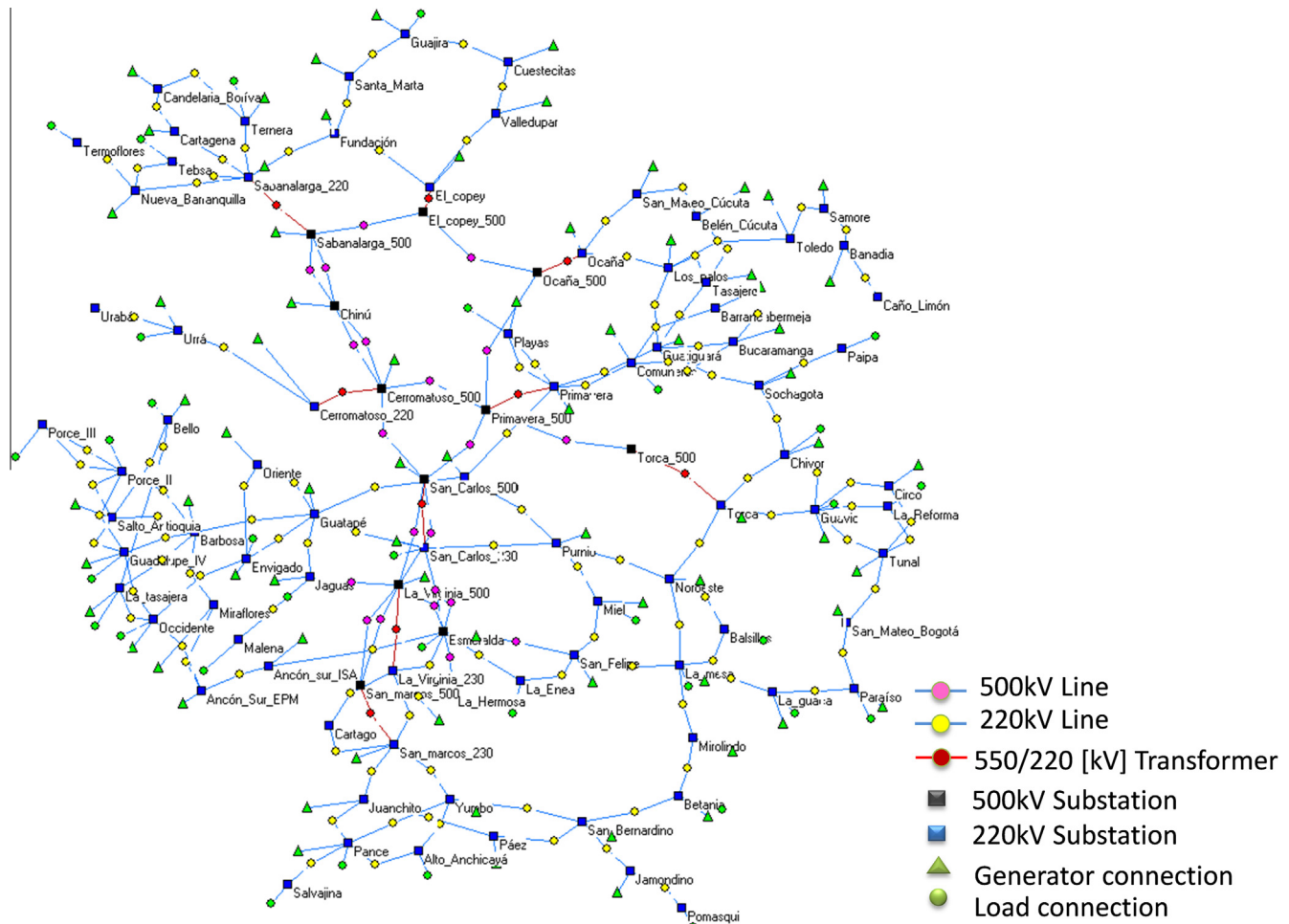


Fig. 4. Scale free graph of 220 kV and 500 kV transmission network in Colombia.

Table 1

Case study on network planning: base case -1-, robustness -2-, expansion -3- and statistic measures on graph representation of power grids from Colombia and Spain.

Country	Transmission network	Case	Circuit length (km)	Buses	Nodes	Links	Average degree ($\bar{k}$)	Graph Average Efficiency ($\bar{e}$)
Spain	400 kV	1	19.622	48	190	212	2.23	0.025
		2		48	198	228	2.30	0.029
		3		76	278	332	2.39	0.027
Colombia	220–500 kV	1	14.300	94	340	380	2.29	0.019
		2		94	348	403	2.32	0.019
		3		117	405	471	2.32	0.025

Table 2

Summary of iterative processes for deliberate attacks and random errors on networks from Colombia and Spain.

Disruption strategy	Network	N° samples	N° iterations per sample
Random	Spain (Case 1)	35	128
Random	Spain (Case 2)	35	131
Random	Spain (Case 3)	35	184
Random	Colombia (Case 1)	35	218
Random	Colombia (Case 2)	35	232
Random	Colombia (Case 3)	35	271
Deliberate	Spain (Case 1)	1	40
Deliberate	Spain (Case 2)	1	42
Deliberate	Spain (Case 3)	1	100
Deliberate	Colombia (Case 1)	1	82
Deliberate	Colombia (Case 2)	1	92
Deliberate	Colombia (Case 3)	1	104

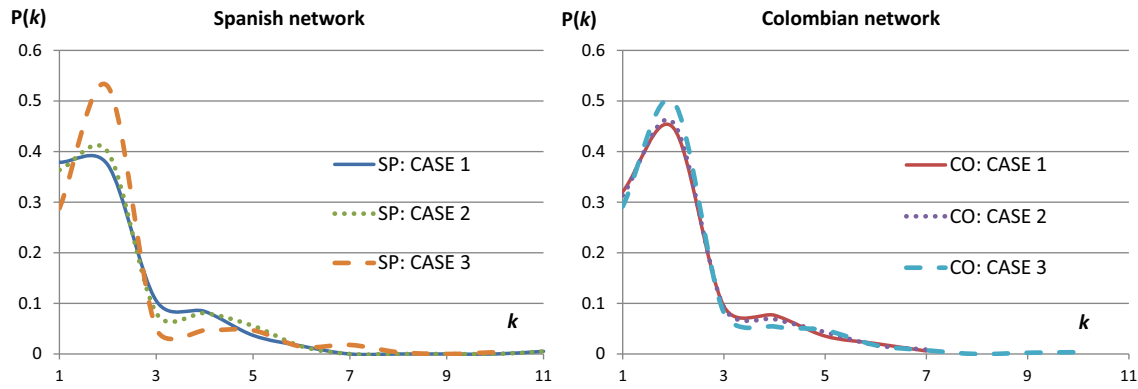


Fig. 5. Comparative results of node degree distribution.

However, 10% of network nodes have degree of connection $k = 3$ (substations on which three transmission lines converge).

The value $k = 2$ is typically related to the graph representation of electric transmission lines (links connecting two substations). The value $k = 1$ refers to those nodes in the graph that have one single connection to other bus bars. They are normally related to power generation or load centres. Fig. 5 also shows that in the Spanish network, the nodal degree distribution $P(k)$ with $k = 2$ increases for case 2 respecting to base case 1. This is consistent since the network expansion in case 2 refers to a more robust mesh by adding new power lines, increasing the proportion of links in the graph with $k = 2$.

Nonetheless, comparing the Spanish network between case 3 and case 1, there is a significant increase on $P(k)$ when $k = 2$. This is explained by growth in 36% of the number of substations and more than 50% in transmission lines as proposed in the expansion plans of the 400 kV network (going from 48 to 76 substations and 212 to 332 links).

Furthermore, comparing the Spanish network between case 2 and case 3, there is slight decrease on $P(k)$ when $k = 3$ and $k = 4$ but slightly increases when $k \geq 5$, which means that new lines and buses preferably connect to major substations.

The results obtained for the Colombian network are very similar to those of the Spanish network in all three cases. However, it can be noted that for the base case 1, the maximum degree is $k = 7$. In case 3 the maximum degree reaches up to $k = 10$. The Colombian power electric grid is less a meshed than the Spanish one (as corroborated from Fig. 5 and from Table 1).

3.4.2. Tolerance to random errors

Recall that the random errors are associated to those risk scenarios of random nature: natural phenomena, inadvertent human error, random technical failure in equipment and hardware, etc. Vulnerability geodesic can be interpreted as the impact on power supply caused by the occurrence of risks that may be associated to disruptions of a percentage f of nodes in the network. As previously explained in 2.2.4, it is correlated to power load shedding of the grid.

The execution of the algorithm for calculating successive contingencies N-1, as described in Section 3.1 originates the results shown in Fig. 6. As a result, the curves of network vulnerability between Spain and the Colombia show that the latter is relatively more vulnerable, since in all cases, for the same fraction of removed nodes, there is a greater impact on the structural vulnerability for the Colombian grid.

A comparison between the case 1 (base case) and case 2 (robustness strategy) reveals a slight improvement of the network vulnerability in both countries. However, comparisons between

case 1 and case 3 (government plans for expansion) show that expanded network is the most vulnerable. An explanation for this has to do with the fact that construction of a less compact network increases the geodesic distance between nodes.

The policies involved into government's plans to build new transmission lines have clear objectives: in Spain they try to attend growing demand, and also to increase the capacity of the electric system in order to transport energy from new renewable generation facilities. Whereas in Colombia those plans seek to provide greater reliability in order to attend growing demand and avoid the isolation of interconnected areas in the event that some disruption event shuts down any region connected to the power grid.

Table 3 shows some data points of node removal effects on isolation and structural vulnerability in the power grid $\bar{v}$, according to tolerance against random errors. This resembles many risk scenarios that are associated with the effect of isolating some geographic areas that may affect a fraction of nodes (5% or 10% of network nodes).

An interesting outcome on this response refers to the effectiveness of those plans to make a more robust network. This can be corroborated when comparing case 1 and case 2, in both countries. Reducing vulnerability $\bar{v}$ in the network is evident when increasing the average value of the degree of connection $\bar{k}$, meaning that if a more meshed network is built with new transmission lines without increasing the number of substations, then a less vulnerable network is expected. Results from Table 3 shows the remarkable effect of decreased vulnerability $\bar{v}$ in the Colombian network for $f \approx 5\%$ and $f \approx 10\%$, and also a slight improvement in the Spanish network.

However, a comparison between case 1 and case 3, which represents the application of expansion plans, shows a possible deterioration of network vulnerability.

Further analysis with the results of Tables 1 and 3, may explain the reason why vulnerability increases in case 3. In Spain, the need to increase the number of buses by 36% (from 48 to 76 buses), but only a growth of 3% in the average nodal degree (from 2.32 to 2.39) conducts to an effect of increasing vulnerability for case 3. Similarly, the growth of 26% required in the Colombian system (from 93 to 117 buses) supports the configuration of a less compact network, but also more vulnerable, since the average nodal degree of the graph only increased by 1% (from 2.29 to 2.32). This means that both systems may cover larger geographic areas, but their robustness (meshing) does not grow at the same rate of their expansion.

3.4.3. Tolerance to deliberate attacks

The vulnerability calculations for deliberate attacks are performed according to a degree based node removal strategy, starting with those that have the most connections. This way, it is possible

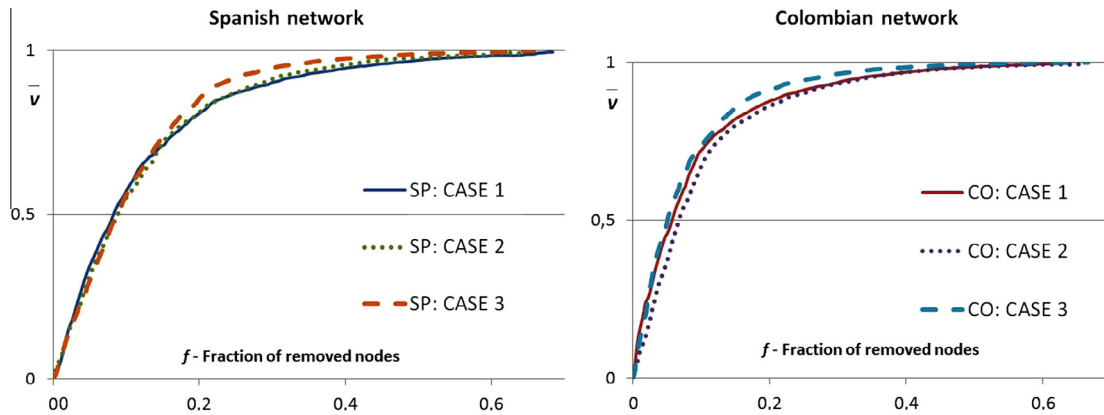


Fig. 6. Random errors – comparison on geodesic vulnerability.

Table 3

Random errors – impact on disconnected users due to vulnerability ($\bar{v}$) as function the fraction of disrupted nodes (f).

Network	$f = 5\%$	$f = 10\%$	$f = 20\%$	$f = 30\%$
Spain (Case 1)	0.34	0.57	0.81	0.90
Spain (Case 2)	0.31	0.55	0.81	0.91
Spain (Case 3)	0.30	0.54	0.86	0.95
Colombia (Case 1)	0.47	0.71	0.88	0.93
Colombia (Case 2)	0.40	0.65	0.87	0.93
Colombia (Case 3)	0.51	0.73	0.91	0.96

to represent risk scenarios that relate to attacks caused by malicious individuals, for instance acts of terrorism, cyber-attacks, acts of vandalism, etc. According to the algorithm showed in Fig. 2, calculation of tolerance against deliberate attacks requires only an iteration to collect results, in contrast to the calculations of tolerance against random errors, which collects 35 sets of results.

Fig. 7 shows the results of structural vulnerability calculation in tolerance against deliberate attacks. As exposed in both cases, the disruption of a small number of nodes ($f < 5\%$) has very high impact on the entire system ($\bar{v} > 85\%$). This means that an attack addressed against substations with high connectivity would represent a shut-down on almost the whole electric infrastructure and therefore a blackout would be expected through a wide geographical region.

The results of vulnerability in Fig. 7 shows a similar behaviour in both networks against deliberate attacks, even though their values are slightly higher in the Colombian network than in the Spanish (for the same fraction of removed nodes).

When comparing case 1 (base case) and case 2 (robustness strategy) for both countries, vulnerability curves practically overlap one over other. This means that the strategy of creating a more

robust network does not imply greater protection against deliberate attacks in infrastructures.

On the other hand, comparisons between case 1 and case 3 (government plans for expansion), show that the expanded network is more vulnerable to deliberate attacks. When building an expanded network, it becomes less compact, and therefore electric power must flow through more paths under node disruption events. This is explained since removal of nodes with higher connectivity has the effect of dramatically increase the geodesic distance between network nodes.

Table 4 shows some data points of node removal effects on isolation and structural vulnerability in the power grid $\bar{v}$, according to tolerance against deliberate attacks.

A comparison between results for both countries in case 1 (base case) and case 2 (robustness) shows no evidence of significant improvements in tolerance against deliberate attacks. Since most of vulnerability values overlap, then the strategy to build a more meshed infrastructure may not be effective enough in deliberate attacks risks.

Particularly in both countries, the expanded network in case 3 is even more vulnerable to deliberate attacks than case 1. This is explained since expansion planning takes into account new connections to substations with high nodal degrees, e.g. substation Madrid (Spain) substation and Esmeralda (Colombia), both with $k = 10$. Additionally, some other buses with $k > 7$ have wide effect on almost the whole power grid when they are disrupted.

4. Discussion

The results presented in Figs. 6 and 7 show that the networks are much more resistant to those scenarios involving random

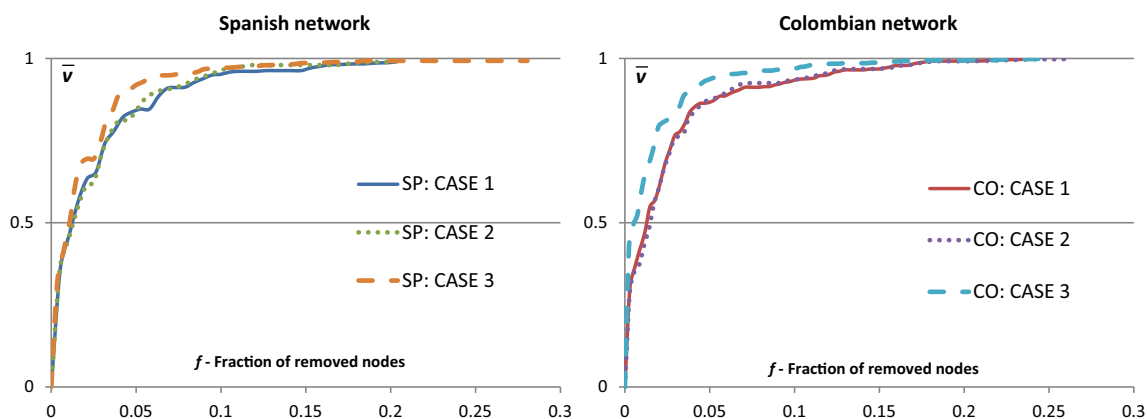


Fig. 7. Deliberate attacks – comparison on geodesic vulnerability.

Table 4

Deliberate attacks – impact on disconnected users due to vulnerability ($\bar{v}$) as function the fraction of disrupted nodes (f).

Network	$f = 0.5\%$	$f = 1\%$	$f = 2\%$	$f = 5\%$	$f = 10\%$
Spain (Case 1)	0.36	0.45	0.62	0.84	0.95
Spain (Case 2)	0.36	0.45	0.62	0.84	0.95
Spain (Case 3)	0.40	0.50	0.69	0.92	0.97
Colombia (Case 1)	0.36	0.45	0.70	0.87	0.97
Colombia (Case 2)	0.36	0.43	0.70	0.87	0.97
Colombia (Case 3)	0.49	0.60	0.79	0.92	0.97

errors risks (natural disasters, technical failures, human errors, etc.) than to those threats related to malicious acts (vandalism, terrorism, cyber-attacks). Given the correlation between load shedding in Eq. (3) and geodesic vulnerability in Eq. (2) it is possible to estimate the impact on the percentage of disconnected users.

Note that those risks associated with random events may result in disruption of a small fraction of nodes. Results for case 1 in the Spanish network show that for random errors with a small fraction of removed nodes, for example $f \approx 5\%$, the impact on the system may isolate $\bar{v} \approx 30\%$ of users (Table 3), but rise up to $\bar{v} \approx 85\%$ in deliberate attacks (Table 4).

As specified in *robustness* plans of case 2 that include construction of new transmission lines, they may be effective only as protection against random error risk scenarios. Deliberate attacks require other protective strategies such as better *resiliency*, that is to say, implementation of methods for network reconfiguration, improving repair times and starting-up operations after deliberate attacks.

In the two analysed infrastructures, expansion plans for the case 3 show an increased vulnerability of the system. This can be explained because the expansion of the transmission grid involves the construction of a larger but less compact infrastructure. The electric power network follows the performance of scale-free graph, which means that a few nodes are highly connected and they have a large number of links to other nodes, although the degree of connection throughout the graph is quite low.

An interesting observation on both networks (Colombia and Spain) is that the expansion plans are conceived for a scale-free graph topology. Networks will grow preferentially on the basis of the nodes of greater connectivity. When those high degree nodes are removed, their converging links are also disrupted and therefore, energy flows in the network have to cross more geodesic paths between the remaining nodes. In the scale-free graph, tolerance on random error implies that when a node is randomly removed, it is very likely to be one of low connectivity degree. In statistical terms, those nodes with less connectivity are the most likely to disrupt.

A comparison of the results presented in Figs. 6 and 7 reaffirm the deduction in which the strategy of improving the *robustness* of the network by building new links between existing buses (case 2) is effective to those risks associated with random phenomena. Nevertheless, the strategy is not quite effective against deliberate threats. On the other hand, addition of new buses into the networks (case 3) is associated with increased vulnerability. This is because the average nodal degree does not increase in the same proportion, since that there are no plans for additional mesh among new added buses.

5. Conclusion

The practical utility of graph theory techniques in power grid systems, relates to the ability of assessing the vulnerability of networks without the need to use electrical parameters. The results presented in this paper support the conclusion that the use of

complex networks, when using appropriate indexes, may be appropriate for the analysis of contingencies.

Through the illustrated application in networks of Colombia and Spain, it has been possible to compare the relative vulnerability between different infrastructure topologies, and the impact of the network expansion plans as set out by governments. The strategy to provide greater robustness to the networks (improvement of the mesh and higher degree of connectivity of buses) provides slight improvement in the vulnerability of the network against random errors (technical failures, natural disasters, human errors, etc.). However, no improvement is evident in the case of deliberate attacks to the infrastructures.

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